Hendrik Nicolas Carlo

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Summary

Computer Science graduate (GPA 3.95/4.00) and Apple Developer Academy graduate specializing in Applied AI, Computer Vision, and ML deployment. Published IEEE researcher with hands-on experience shipping production iOS applications and scalable ML inference systems. Proven ability to bridge research-grade models and real-world mobile deployment using CoreML, Swift, Docker, and cloud infrastructure.

Education

- **Bina Nusantara University, School of Computer Science** — Jakarta, Indonesia
  Bachelor of Technology - Information Technology; GPA: 3.95/4.00 — *Sept 2022 – Feb 2026*
  - **Relevant Coursework:** Analysis of Algorithms, Data Structures, Artificial Intelligence, Machine Learning, Computer Vision, Software Engineering, Operating Systems, Database Systems.

Experience

- **Apple Developer Academy @BINUS** — Tangerang, Indonesia
  Junior AI Engineer & iOS Developer (Internship) — *March 2025 – Dec 2025*
  - **Product Development & Deployment**: Developed **5 iOS applications** in 10 months, resulting in **2 App Store launches** and **2 TestFlight betas**. Applied the **Challenge Based Learning (CBL)** framework to drive iterative product design, validation, and deployment.
  - **Lead the AI development of Rulaa, a Clothing Size Recommendation & Body Measurement App**: Architected and deployed serverless ML inference pipelines using **Docker**, reducing latency for model serving. Built the iOS frontend in **SwiftUI/UIKit** and **Vision** using MVVM architecture and secure REST APIs.
  - **Leadership & Research**: Founded and led the internal **Machine Learning Club**, delivering technical workshops on the **Apple AI ecosystem**. Prototyped **2 macOS applications** leveraging **CoreML**, **Vision**, and **CreateML** to demonstrate production-ready on-device intelligence systems.
- **Apple Developer Institute for AIML @ UC** — Surabaya, Indonesia
  AI/ML Engineer & iOS Developer (Full-time) — *April 2026 – Present*
  - **Applied AI/ML research and engineering**: Built and evaluated ML systems that run fully offline on consumer hardware, such as on-device LLM agents, 3D mesh reconstruction, and a trading agent. Recurring approach: domain structure and deterministic scaffolding in place of larger models.

Publications

- **Comparative Analysis of Fungal Infections Classification in Apple Leaves**
  Published in *6th International Conference on Cybernetics and Intelligent System (ICORIS), IEEE, 2024*
  - Co-authored a paper comparing standard CNNs vs. CNNs with texture analysis (GLCM), providing critical analysis on feature extraction efficiency.
  - **Authors: Carlo, H. N.**, Andreas, K., & Zakiyyah, A. Y.
- **JKTDrive: Early Explorations of Visual-Language Model on Autonomous Driving in Indonesia**
  Accepted in *12th International Conference on Computing and Artificial Intelligence (ICCAI), IEEE, 2026*
  - Co-authored a paper Benchmarking VLM performances as a reasoning model for Autonomous Driving Agent in Indonesia.
  - **Authors: Carlo, H. N.**, Sanjaya, J. S., Tanjaya V., & Cenggoro, T. W.

Achievements

- Top 20 Microsoft AI for Accessibility - May 2024
- Semi-finalist Samsung Solve for Tomorrow - Aug 2025

Skills Summary

- **Languages**: Python, Swift
- **AI & Data**: Transformers, TensorFlow, PyTorch, Keras, Scikit-Learn, Pandas, NumPy, OpenCV, CoreML
- **Backend & Tools**: FastAPI, Docker, Git, AWS, GCP
- **Soft Skills**: Challenge Based Thinking, Technical Writing, Research, Public Speaking, Agile Leadership